

Suitability of using the global positioning system (GPS) for studying Feral Pigeons *Columba livia* in the urban habitat

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Capsule GPS tracking gives very precise information about Feral Pigeons' spatio-temporal behaviour in the urban habitat.

Aims To test the suitability and the limits of GPS tracking in the urban habitat for a detailed analysis of Feral Pigeons' spatio-temporal behaviour.

Methods We placed ten receivers in eight different locations in the city of Basel, Switzerland. Between 1 and 23 April 2003, we performed 166 recordings and compared the stored positions with the real location. We also tested the GPS receivers on 29 free-living Feral Pigeons.

Results Almost 82% of the positions obtained with the GPS receivers were within 25 m and 96% within 100 m of the real location. The accuracy varied between locations, depending on the proportion of open sky. In 38% of the tests, no positions were stored. We performed 143 test flights with 29 Feral Pigeons (18 males and 11 females). A total of 118 flights produced 'storable' position information, 25 flights (17.5%) produced no storable data. Over 47% of the flights were complete (beginning and ending at loft), the others began or ended elsewhere. We encountered some difficulties: delays to get the first fix; reflection of the satellite signal on tall buildings; and limited battery life.

Conclusion Despite some difficulties related to the urban habitat and the technical features of the GPS receivers, we recommend the GPS-based tracking method for studying the spatio-temporal behaviour of Feral Pigeons and other birds weighing over 300 g and which are easy to capture.

The daily activity rhythms and the home range of Feral Pigeons are not well understood, despite a great number of studies involving a variety of methods. Methods used include: (a) direct observation of pigeons commuting to the surroundings or flying within the city (Janiga 1987), (b) observation of individually marked birds at different places in town (Sol & Senar 1995), (c) use of an automated system based on electronic rings detected by an antenna at feeding places (Dell'Omo 1997) and (d) telemetry (Scholl & Häberling unpubl. data). Each method is limited, leading to sampling bias dictated by the observation points and time needed for observation. Direct observation gives a first indication of where feeding occurs, but it is difficult to make statements about the individual foraging strategies. Marked individuals are sometimes difficult to locate, because pigeons can fly to places

where recording is not possible (e.g. inner courts of private buildings). Sometimes the markings are difficult to recognize, e.g. when the birds are sitting too far away or because they escaped when the observer came too close. With an automated system the pigeon is registered only in places where the antennae are placed. Telemetry needs to follow the bird during observation and this can be difficult in towns because of the complex spatial structure.

A GPS (global positioning system) has been used successfully to monitor the flight of homing pigeons from the release point to the home loft (Von Hünenbein *et al.* 2000). The results of these studies were gathered under optimal conditions: in open field habitat without disturbance by high buildings and trees; easy handling by using tame homing pigeons; the pigeons flew directly back to their home loft, so the battery life in the GPS receivers was not a constraint. Weimerskirch *et al.* (2002) tracked foraging Wandering

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Albatrosses *Diomedea exulans* using GPS under good receiving conditions in open sea.

We set out to test the suitability and limits of GPS tracking in the urban habitat to perform detailed tracking of bird movements. We expected to encounter two main problems (Garmin Corporation 2000a): (a) reflection of the signals on buildings and (b) insufficient satellite signals, when the proportion of open sky is too restricted. Reflection can result in recording of shifted positions that are several hundred metres away from the real position. Limited reception of signals can lead to interruptions in the record, if there are not enough satellites available to calculate the position.

MATERIAL AND METHODS

GPS receivers

The GPS system is based on a network of satellites that continuously transmit coded information. It allows precise location information to be gained by measuring the distances from the earth to the satellites (Mehl 1996). A GPS receiver stores its own position calculated from its distance to at least four satellites every second (continuous mode) or at another time interval chosen by the user (low-power mode). The stored data are downloaded from the GPS receiver to a PC where the positions are analysed and represented graphically on a map.

For our tests, we used ten GPS receivers (SAM, GPS Smart Antenna Module, basing on the TIM module) designed by u-blox AG (Thalwil, Switzerland) and CabTronix GmbH (Kloten, Switzerland) (for technical features see u-blox AG 2003). Our receivers were 60 × 32 × 14 mm in size and weighed between 29 and 36 g, depending on the size of the battery. The GPS receiver is 10–15% of a Feral Pigeon's body weight. When equipped with a 960-mA battery (Li-poly accu SLPB523462 with PCM, Worley, Australia), recording lasted about 5 h in continuous mode or at least 54 h in low-power mode (with the setting of one position every 5 seconds). Steiner *et al.* (2000) discuss technical considerations and limitations of GPS tracking.

There are three options for data storage and retrieval (Rodgers *et al.* 1996): (a) storage of all data onboard, with retrieval upon recapture of the animal, (b) storage of data and transmission for retrieval by secondary low earth orbit satellite link or (c) storage of data and transmission to a local computer by radio link. We chose

option (a), because the other two need supplementary modems for communication, which would increase the weight of the GPS receivers.

We used the software μ -logger (u-blox AG, Thalwil, Switzerland) for downloading. Positions from the GPS system are given in WGS-84 coordinates (World Geodetic System 1984, an earth-fixed global reference frame). We transformed these into CH-1903 coordinates (Swiss geodetic coordinates) and projected them on the electronic map of Basel (from the Grundbuch- und Vermessungsamt, Justizdepartement des Kantons Basel-Stadt), to show each position of the pigeons. The stored positions were represented on the electronic map by the software MapInfo Professional (MapInfo Corporation Troy, New York).

Accuracy tests

To test the accuracy of the GPS positions obtained in an urban area, we placed stationary GPS receivers at different locations in the city of Basel, Switzerland (see Table 1). The satellite geometry (the satellites' positions in three-dimensional space) greatly influences the location accuracy of GPS positions (Dussault *et al.* 2001). The ability to determine a position deteriorates if the four satellites used to take measurements are close together (Zogg 2002). If there are more than four satellites available, the GPS receiver chooses those that give the best satellite geometry. In urban areas, access to open sky is restricted and this often results in poor satellite geometry. Our GPS receivers do not indicate the satellite geometry they used to calculate the positions. We therefore tested the reliability of GPS position recording at least twice at exactly the same location at different times so as to receive different satellite geometries (the satellites are not stationary). Eight different locations with a different amount of open sky space were determined for that purpose. The locations were classified in categories according to the proportion of open sky.

We placed between six and ten GPS receivers simultaneously at one location and left them stationary for at least 32 min (maximum 101 min). Half of the GPS receivers were running in continuous mode (79 records) and the others (87 records) were in low-power mode set on one fix every 3 s. We configured the GPS receivers to store only positions calculated with four or more satellites. When navigating with less than four satellites the GPS utilizes the last computed altitude (Garmin Corporation 2000b). With stationary GPS receivers this can achieve good results (u-blox AG

2002) but with pigeons that rapidly change altitude the positions would be less accurate.

We compared the positions stored by the GPS receivers to the real positions using the electronic map of Basel. We observed that some stored positions were on altitudes that are not possible in Basel (e.g. below the lowest altitude of Basel). In such cases, the horizontal position error may be as large as the altitude error, e.g. when calculations are based on bad satellite geometry. Subsequently, we developed a filter to reject all non-sense positions (altitudes below 250 m and above 390 m).^a We compared this new set of data with the real positions to verify the reliability of our altitude filter.

We also measured other indicators of data quality: the time-to-first-fix, i.e. the time between the onset of receiver operation and the time of the first recorded position; the number of interruptions lasting longer than 1 min and longer than 5 min; the maximum duration of interruption; and the rate of failure (no stored positions during the record). We tested the differences between the receivers, the test locations and the operating modes using these indicators of data quality (see Table 2).

Tests with the pigeons

The pigeons used for this study were living in three lofts situated in public buildings in Basel, Switzerland. The loft of the Matthäus church has a surface of 27.8 m² and contains 39 breeding boxes. Around 90 pigeons were resident in this loft. The Stapfelberg loft is 22 m² and was used by around 35 pigeons. The loft of St Peter's church is 11 m² and around 15 pigeons were resident there. The birds foraged in the city without any supporting food or water in the loft. The pigeons, therefore, show behaviour and activity patterns typical of free-living urban Feral Pigeons. We studied 11 females (average weight 339 ± 22 g) and 18 males (average weight 354 ± 25 g) known to be closely attached to the lofts, since they were breeding during the experiment or had bred a few weeks before. We studied only subjects in good body condition (birds that weighed over 300 g). We checked their weight repeatedly during the training and the subsequent GPS experiment.

In the Matthäus loft, we constructed special breeding boxes that can be closed from outside the loft to catch the breeding birds. Pigeons sitting outside the boxes and those in the other lofts were caught in the lofts with a net.

Before releasing the pigeons equipped with GPS receivers, they were accustomed to carrying dummies of

the same size and weight as the receivers for four to nine days. None of the birds seemed to be handicapped by the dummies, so all trained pigeons were used for the experiment. Only immediately after fixing the dummy on the birds, did we observe intensified preening, which stopped after a few minutes. Consequently, we believe that the GPS receivers do not irritate the pigeons or significantly influence their behaviour.

Both the dummies and the GPS units were fixed onto the pigeons' back with Velcro tape glued to the feathers with cyanoacrylate (power-glue). To ensure a good fixation of the Velcro, the feathers were cut down to approximately 3 mm. Since some pigeons lost their dummies at the beginning of the study, we developed a supplementary fixation with a lightweight harness of elastic ribbon. Two loops of the ribbon passed around the pigeon's body, one anterior to the wings, the other between the cloaca and the legs. The two loops were connected on the abdomen with a strip that was sewn directly with silk to adapt it individually to the size of each pigeon. The GPS receivers were covered with a plastic film during the flights to protect them from precipitation and soiling.

We caught the pigeons in the morning to fix the GPS receivers onto their backs and released them immediately in the loft. The records started when the pigeons flew out of the loft, since there is no reception of satellite signals inside buildings. There was always a delay between leaving the loft and starting the record, due to the acquisition time of the GPS receiver (typically 45 s, u-blox AG 2003). In the evening, we removed the GPS from the homed birds to charge the battery and download the data to the PC. Between 12 February and 31 March 2003, we performed 143 test flights in low-power mode with 29 Feral Pigeons (18 males and 11 females).

RESULTS

Accuracy tests

We obtained 166 records from ten unmounted GPS receivers at eight different places in the city of Basel. One receiver gave poor results due to problems with the battery so we eliminated this from the study. For 96 (61.9%) of the remaining 155 records, the GPS receivers successfully stored data.

Horizontal accuracy

Over 81% of all stored positions were within 25 m of the real location and 96.3% within 100 m. The

Table 1. Accuracy of the recorded positions at each test location (category 1: 76–100% of open sky; category 2: 51–75%; category 3: 26–50%). For each location, we give the total number of records performed, the number of successful records (with positions), the mean percentage of positions within 25 m and 100 m of the real position, the maximum distance error (both for uncorrected and corrected data set) and the vertical location error.

Location	Proportion of open sky, category	No. of records			Mean % of fixes				Max. horizontal location error (m)		Mean vertical location error (m)
		Total	With positions	No. of positions	Within 25 m	Corrected data set	Within 100 m	Corrected data set	Uncorrected data set	Corrected data set	
Mittlere Rheinbrücke	1	14	12	13 070	95.33	95.33	100	100	86	86	45
Johanniterbrücke	1	18	16	26 730	90.32	94.65	95.48	99.7	564	207	52.9
Matthäuskirche	1	18	11	35 198	86.63	88.17	99.53	100	198	77	57.8
Dreirosenbrücke	1	18	14	18 011	86.83	92.47	95.88	100	280	96	–
Stapfelberg	2	14	11	24 180	85.81	89.43	100	100	85	72	46.3
Harbour St Johann	3	18	11	7022	41.97	52.37	85.94	100	252	64	96.8
Marktplatz	3	32	12	21 131	79.57	90.24	96.61	99.99	199	103	44.2
Petersplatz	3	23	9	9661	87.7	90.08	96.71	99.87	630	113	52.4
Total or mean		155	96	155 003	81.77	86.59	96.27	99.95	286.75	102.25	56.49

accuracy of the GPS positions varied between the different test locations (Table 1). The mean percentage of positions within 25 m of the real location differed between the test locations (minimum 42.0% and maximum 95.3%). The mean percentage of positions within 100 m varied between 85.9% and 100%. We measured horizontal location errors (i.e. the distance between the real location and the measured position) up to 630 m.

After the elimination of positions with large altitude errors (under 250 m and over 390 m asl), the mean percentage of positions within 25 m of the real location increased from 81.8% to 86.6% (Table 1). The mean percentage of positions within 100 m was also higher: >99.9% instead of 96.3%. The maximum horizontal location error was reduced to 207 m after the data were corrected.

There was no statistically significant difference in the horizontal accuracy (percentage of positions within 25 m of the real location) between the receivers (three-way-ANOVA, $P = 0.34$, $n = 96$). The records in continuous mode gave slightly more accurate positions and approached significance ($P = 0.06$). There was a significant difference between the categories of locations ($P = 0.0009$), but this difference was only due to one test site, the harbour St Johann, that differed considerably from the other locations ($P < 0.0001$). When testing the difference between the categories of locations excluding this problematic test site, the results were still slightly better in category 1, which had the more accurate positions, and category 2 compared with category 3, but the difference is not significant (three-way-ANOVA, $P = 0.34$, $n = 85$).

Vertical accuracy

The mean difference between the altitudes calculated by the GPS and the real altitudes was 56.5 m (Table 1). The smallest difference was 44.2 m and the greatest difference 96.8 m. We made no further analysis on the vertical accuracy, since the height calculated by GPS receivers is not directly comparable to altitudes above sea level.

Other indicators of good quality data

Table 2 shows the values obtained for each indicator of data quality at the different test locations. The interruptions and the long duration of time-to-first-fix occurred when the GPS receiver was unable to get signals from at least four satellites.

The statistical tests (three-way-ANOVA, logistic regression and Poisson regression for the different indicators of data quality) showed no significant differences between the receivers.

The recordings in continuous mode gave significantly shorter interruptions than in low-power mode (three-way-ANOVA, $P = 0.01$, $n = 96$). The continuous mode also had fewer interruptions. The difference was statistically significant for the number of interruptions >1 min (Poisson regression, $P = 0.02$, $n = 96$) but not for the interruptions >5 min ($P = 0.54$). In those cases where we included the harbour, the difference in the maximum duration of interruption between the location categories was significant ($P = 0.008$) but when we excluded the harbour the maximum interruption was no longer significantly shorter in categories 1 and 2 compared with category 3 ($P = 0.75$, $n = 85$). The rate of failure was also significantly greater in location

Table 2. The ease of obtaining positions at each location, expressed as the time needed to get the first position (time-to-first-fix) and the number of interruptions that occurred during the test. The total operating time is the number of minutes between the first fix and the end of each test summed for all tests in each location.

Location	Time-to-first-fix (min)			No. of interruptions >1 min	No. of interruptions >5 min	Maximum interruption (s)	Total operating time (min)
	Mean	Min.	Max.				
Mittlere Rheinbrücke	4.5	<1	19	0	0	19	359
Johanniterbrücke	5.75	<1	25	4	0	124	678
Dreirosenbrücke	5.5	1	26	11	2	300	516
Stapfelberg	5.3	<1	33	6	1	485	628
Harbour St Johann	16.5	<1	38	30	13	2340	480
Marktplatz	9.5	<1	29	9	0	223	556
Mathäuskirche	7.2	<1	29	44	11	2410	1019
Petersplatz	34	16	55	16	1	636	245
Mean	11.0	8.5	31.75	15.0	3.5		4481

category 3 (with or without the harbour) compared to location category 2 (with harbour $P = 0.013$, $n = 155$, without harbour $P = 0.004$, $n = 137$) and to category 1 ($P < 0.0001$ with or without the harbour). The other indicators of quality data gave no significant differences between the location categories or between the operating modes.

Tests with pigeons

Of the test flights with pigeons, 118 produced storable position information and 25 (17.5%) produced no stored data (Table 3). There were 56 (47.5%) complete flights and the others were incomplete (Table 3); 14 flights (11.9%) had a record duration of less than 1 h. There were no significant differences between the number of successful flights obtained with males or with females (Fisher's exact test, $P = 0.37$, $n = 143$), the number of complete flights ($P = 0.61$), the number of flights beginning at the loft ($P = 0.21$), ending at the loft ($P = 0.68$), neither beginning nor ending at the loft ($P = 1.0$) and the number of flights lasting less than one hour ($P = 0.78$). The pigeons always returned to the lofts in the evening except for three flights: two pigeons

returned the next day and one pigeon returned two days later.

We monitored the pigeons' body weight during the entire experiment. After the training with the dummies (4–9 days), the pigeons had significantly lost weight (one-sample- t -test, $P = 0.003$, $n = 29$). The mean loss was 2.6% of the body weight. The females showed a significantly greater weight loss (4.6%) than the males (1.5%, Wilcoxon–Mann–Whitney test, $P = 0.04$, $n = 29$). We employed mixed linear models to examine if there were changes in body weight after the training, but there were no more significant changes.

DISCUSSION

Accuracy test

In our tests, 81.8% of the positions were within 25 m of the real location and 96.3% within 100 m. Elimination of obviously false positions by altitude resulted in higher percentages (86.59% and 99.95% respectively). Theoretically, 95% of the time an accuracy of ≤ 13 m in the horizontal and ≤ 22 m in the vertical plane is

Table 3. Flights performed by male and female Feral Pigeons carrying GPS receivers. Flights of less than one hour were not included in the analysis.

Sex	No.	Number of flights (%)						Flight duration (min)	
		With data	Complete	Beginning at loft	Ending at loft	Not beginning or ending at loft	Lasting < 1 h	Mean	Range
Male	82	70 (85.4)	34 (48.6)	4 (5.7)	18 (25.7)	5 (7.1)	9 (12.9)	286	75–522
Female	61	48 (78.7)	22 (45.8)	7 (14.6)	11 (22.9)	3 (6.3)	5 (10.4)	336	62–566
Total	143	118 (82.5)	56 (47.5)	11 (9.3)	29 (24.6)	8 (6.8)	14 (11.9)	307	

attained (Zogg 2002), assuming optimal conditions with completely open sky and good satellite geometry. Dussault *et al.* (2001) performed tests in open areas: 95% of the positions were within 250 m and 50% within 160 m. They performed their study before removal of the selective availability (the United States Department of Defense previously intentionally degraded the signals transmitted by the GPS satellites for civilian users). Dussault *et al.* (2001) applied a data correction eliminating positions obtained with bad satellite geometry, increasing the accuracy to 95% within 75 m and 50% within 15 m. The accuracy of the GPS positions in our tests is comparable to the results obtained by Dussault *et al.* (2001).

The maximal horizontal location error we measured was 630 m. It is in the same range as the 650 m obtained by Rempel *et al.* (1995) under boreal forest canopy.

The accuracy of positions obtained during the flights with pigeons can hardly be verified. We expect the positions to be more accurate since pigeons spend a lot of time on the roofs of buildings, where the proportion of open sky is nearly unrestricted. But gaps in the record are inevitable when the pigeons fly down to the streets to feed or when sitting on the walls of buildings.

We made no further analysis of the vertical accuracy, since the height calculated by GPS receivers is not directly comparable to altitudes above sea level (see section 'Accuracy tests' in Material and methods).

We found no statistically significant differences in the quality of data obtained with the different receivers. Therefore, we conclude that the differences that occurred are due to the operating mode and the locations. We obtained more accurate positions and less interruptions with receivers operating in continuous mode. Since we intended to record the pigeons' flight activity for the entire day, we had to set the receivers in low-power mode. The accuracy of GPS positions in this operating mode could be further improved with the use of differential GPS (a costly system that uses the known position of a local reference station to correct the obtained GPS positions).

Problems with the urban habitat

We expected to encounter two problems connected to the structure of the urban habitat: reflection of the satellite signals on high buildings and insufficient satellite signals. During the accuracy tests, at the harbour St Johann location (characterized by high

grain storage buildings) we achieved significantly less precise data than at all other locations. Without this problematic location, the categories of locations differed significantly only in the rate of failure. Generally, the accuracy of the results depended on the degree of open sky, but the differences were not statistically significant.

Some flight paths of pigeons showed positions that we interpreted as shifted positions because of the reflection of satellite signals. With the accuracy tests we proved that reflection had occurred.

The lack of sufficient satellite signals occurred when the records had interruptions or if there were long times-to-first-fix. Rempel *et al.* (1995) had similar problems under forest canopy (at some locations only 10% of the attempts to get a position were successful). We had comparable results in our tests: the time-to-first-fix varied among the different locations, being shorter at open locations than on narrow public squares. The interruptions during the records were also more frequent at locations with a smaller percentage of open sky. The positions stored immediately before and after interruption indicate where these interruptions occurred and if the pigeon moved during the gap. During flights, the access to open sky is better, so we expect the record to restart quickly if the pigeon flies during a gap.

Problems with the behaviour of Feral Pigeons

Feral Pigeons are not really used to handling by man and they are free-ranging (Haag-Wackernagel 2000), so their behaviour is unpredictable. The GPS-based method requires the return of the carrier bird to the loft or to a site where the receiver can be removed. Our tests showed that the trained pigeons always returned to the loft in the evening or the following day, even after some weeks of daily disturbance by human activity in the loft. Handling, and especially capturing, represents stress for a wild bird. This might influence normal behaviour. After handling, however, the Feral Pigeons from our lofts returned rapidly to their normal activity, e.g. breeding birds returned to their eggs, as we observed in the loft. Consequently, we believe that the GPS receiver does not irritate or disturb the pigeons significantly.

The pigeons showed an average weight loss of 2.6% at the beginning of the tests during the training with the dummies, but afterwards the weight stabilized. For comparison, we calculated an intraindividual standard deviation of 8% in the weight variation in a data set

obtained over a few years with undisturbed Feral Pigeons in Basel ($n = 28$). During training, females showed a significantly greater weight loss than males. Despite this, females did not behave differently during flights with the receiver (no significant differences in the numbers of successful flights, complete flights or incomplete flights).

Technical problems

The GPS receivers needed a few minutes to store their first position (time-to-first-fix), especially when the receivers were at a location with poor satellite reception. Even under optimal conditions the delay due to the acquisition time required by the receiver is around 45 s. This can lead to incomplete records.

The battery life may be insufficient to record the entire day's activity if the pigeons are equipped with the device in the morning but fly out to forage only in the late afternoon. During their stay in the loft, the GPS receiver continuously tries to get satellite signals. This energy-consuming process has limiting effects similar to using continuous working mode. The capacity can be increased only by adding a heavier battery or by lowering the sampling frequency (low-power mode). A weight increase would, however, be problematic. We think that the present weight already represents the maximum a pigeon can carry without influencing normal behaviour. Kenward (2001) indicates that adverse effects tend to emerge in the long-term for harness-mounted tags weighing more than 4–5% of body mass. Gessaman & Nagy (1988) and Gessaman *et al.* (1991) tested the flight performances of homing pigeons and tippler pigeons with a load of 5% of the body mass. The load decreased the speed and increased the intensity of the pigeons' metabolism during long flights, but the pigeons nevertheless remained able to perform long flights (320 km). In our tests, the pigeons carried GPS receivers representing 10–15% of their body mass, but they did not have to fly over long distances and did not carry the receivers for a long time.

The sampling frequency can be lowered to save power, but the accuracy of the positions consequently decreases (u-blox AG 2002). Another way to save power is to set longer standby phases when the receivers cannot get satellite signals. But the resulting information loss may be crucial.

The results show that 17.5% of flights had no stored position. The failures can be explained in three ways: either the pigeon leaves the loft with an empty battery,

the pigeon does not leave the loft at all or the pigeon flies out and spends all the time at a place where the satellite signals cannot be recorded (not enough open sky, poor satellite geometry).

Our tests with GPS receivers confirm that GPS tracking is an appropriate method for studying Feral Pigeons, despite some problems due to the structure of the urban habitat, the technical features of GPS receivers and the unpredictable behaviour of wild birds. The stored data can be filtered (e.g. according to altitude) to eliminate inexact positions. Therefore, very precise information about a bird's spatio-temporal use of a habitat can be supplied. Some elements of a species' lifestyle are required to render it suitable for GPS tracking: with the actual weight of the GPS receivers, the animal's body weight should be over 300 g, and they must be easy to capture and recapture. In the urban habitat, recording is difficult on the ground of the street canyons. Therefore, GPS tracking is unsuitable for animals staying only at ground level.

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ENDNOTES

a. To develop the altitude filter we determined the altitude range within which the pigeons were expected to stay. The lowest altitude in Basel is the Rhine river (240 m asl). We recorded pigeons on the top of the highest buildings (grain storage building in the harbour St Johann, 315 m), but they

never flew at high altitudes above these high buildings. Our accuracy tests revealed that the altitudes calculated by the GPS receivers are always higher than the real altitudes (mean 56 m, see Results). The real altitudes are defined here as altitudes in Switzerland measured on sea-level of the Mediterranean Sea. The height determined by GPS measurements relates to the perpendicular distance above the reference ellipsoid and therefore differs from the height datum Mean Sea Level (MSL). The difference between altitudes approximated by the WGS-84 ellipsoid and the MSL measurements can come up to 100 m (u-blox AG 1999). To avoid excessive information loss, we set an altitude filter to reject GPS altitudes below 250 m and above 390 m, including margins above and below the pigeons' normal altitude range and also the difference between GPS height and altitude above sea level.

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